MILITARY SPECIFICATION

PROJECTILE, 40MM, XM577

1. SCOPE

1.1 This specification is the applicable detail specification for Projectile, 40mm, XM577.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-P-10025	- Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

Munitions Command

B10542354	- Projectile, 40mm, XM577
IEL-10542354	- Index of Inspection Equipment Lists for Projectile, 40mm, XM577

FSC 1310

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications, and the following:

3.2 First article sample.- Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

3.3 Plastic defects.- All components and assemblies shall be free from blisters, bubbles, cracks, crazing, mottle, porosity or any other defects.

3.4 Sabot base integrity.- The base of the sabot shall show no evidence of failure or permanent deformation when the applicable force is applied in an axial direction.

3.5 Functional.

3.5.1 Temperature condition.- The projectile when assembled into complete rounds shall be temperature conditioned at +20, +70, and +140 degrees F. for a minimum of twelve hours and tests for the following phases:

3.5.2 Target pattern.- The projectile assembly when assembled into complete rounds and fired, shall not fail to release pellets prior to target impact and shall produce a target pattern at 40 meters containing no less than 13 pellets within a 5 foot diameter.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Burr.- No part shall have a burr which might interfere with the assembly or function of the round, or which might be injurious to personnel handling the item.

3.6.2 Foreign matter.- No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 120 projectile assemblies as follows:

100 projectile assemblies, complete
10 sets of subassemblies
10 sets of component parts unassembled

Ninety of the above assemblies shall be forwarded to a designated proving ground for ballistic testing. The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

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4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or MIL-STD-1235, as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed by MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Cap, Drawing B10542345, covering a detail of B10542354.

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Material defective	Visual
MINOR		
201.	Foreign matter	Visual
202.	Marking missing, incorrect or illegible	Visual

4.3.2.2 Cup, pellet, Drawing B10542360, covering a detail of Drawing B10542354.

CRITICAL		
	None defined	
MAJOR		
101.	Diameter of flange	Gage
102.	Diameter of counterbore	Gage
103.	Thickness of flange	Gage
104.	Location of dimples	Gage
105.	Dimples missing or incorrect	Visual (see 6.1)
106.	Material defective	Visual
107.	Flash on flange	Visual

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4.3.2.2 (cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter, front of cavity	Gage
202. Diameter, rear of cavity	Gage
203. Diameter, rear of flange (outside)	Gage
204. Diameter at rear (outside)	Gage
205. Depth from front face to bottom of cavity	Gage
206. Depth of dimples	Gage
207. Length total	Gage
208. Radii of chamfers missing or incorrect	Visual (see 6.1)
209. Foreign matter	Visual

4.3.2.3 Sabot, Drawing C10542356, covering a detail of Drawing B10542354.

CRITICAL	
None defined	
MAJOR	
101. Diameter rear of flange	Gage
102. Depth of center cavity	Gage
103. Drag slots missing	Visual
104. Flange missing	Visual
105. Stop pad missing	Visual
106. Material defective	Visual
MINOR	
201. Diameter at front of center cavity	Gage
202. Diameter at rear of center cavity	Gage
203. Diameter of obturator boss	Gage
204. Length of stop pad	Gage
205. Length of obturator boss	Gage
206. Marking missing, incorrect or illegible	Visual
207. Excessive flash	Visual
208. Foreign matter	Visual

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4.3.2.4 Obturator, Drawing C10542355, covering a detail of Drawing B10542354.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter of hole	Gage
102. Diameter at large end (outside)	Gage
103. Thickness at base	Gage
104. Material defective	Visual
MINOR	
201. Length total	Gage
202. Excessive flash	Visual (see 6.1)
203. Radii or chamfers missing or incorrect	Visual (see 6.1)
204. Burr	Visual
205. Foreign matter	Visual

4.3.2.5 Pellet cup assembly, Drawing B10542359, covering a detail of Drawing B10542354.

CRITICAL	
None defined	
MAJOR	
101. Weight	Approved scale
102. Improper assembly	Visual and Manual
MINOR	
None defined	

4.3.3 Inspection testing.- The following tests shall be performed in accordance with 4.4.

4.3.3.1 Cap.

4.3.3.1.1 Push test.- 100 caps from each production lot of caps shall be subjected to this test to determine compliance with the applicable drawing. Complete separation prior to application of the minimum force or failure to completely separate at the maximum force shall be cause for rejection of the production represented by the sample.

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4.3.3.2 Sabot base integrity.- Five sabots shall be selected from each eight (8) hours production shall be subjected to this test to determine compliance with 3.4. Failure of any sabot to comply will be cause for rejection of the lot.

4.3.4 Functional.- The following proving ground tests shall be performed by the Government.

4.3.4.1 First article sample.

4.3.4.1.1 Initial production sample.- The quantities listed in Table I will be ballistically tested at the phase indicated.

TABLE I

PHASE	QUANTITY
Target Pattern	90

4.3.4.2 Production lot sample.- The quantities listed in Table II will be ballistically tested at the phase indicated.

TABLE II

PHASE	QUANTITY
Target Pattern	60

Failure of any projectile to comply with 3.5 shall be cause for rejection of the lot.

4.3.5 Inspection equipment.- Index of Inspection Equipment List No. IEL-10542354 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL. (Reference is made to Inspection Equipment List for definition of coding format.)

4.4 Test methods and procedures.

4.4.1 Cap.

4.4.1.1 Push test.- The samples supplied in accordance with 4.3.3.1.1 shall be mounted on an approved device in such a manner that the device is held stationary. The force will be applied to the cap until the minimum required force is reached. If complete separation has not occurred, the application of the force will be continued until completed separation has occurred or the maximum required force has been attained.

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4.4.2 Sabot base integrity.- The sample supplied in accordance with 4.3.3.2 shall be mounted in an approved device in such a manner that the sabot will be held stationary. A 2000 pound force will be applied to the base of the center cavity. After the force has been released, the sabot will be visually examined for failure to comply with 3.4.

4.4.3 Functional.

4.4.3.1 Target pattern.- The projectiles supplied in accordance with Table I and II will be assembled into complete rounds and fired in a launcher for which the round is standard. The weight of the propellant charge will be that used in service firing. A target measuring 8 feet by 8 feet shall be placed 40 meters from the muzzle of the launcher. Observations shall be made to determine compliance with 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 Level A.- Not applicable

5.1.2 Level B.- Not applicable

5.1.3 Level C. - Packing and marking shall be in accordance with MIL-P-10025.

5.2 Data card.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characterisitc may be measured or gaged to determine acceptability.

6.2 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

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6.3 Combining of proving ground test.- When the contractor for the cartridge is also the contractor for one or more of the components thereof, the proving ground tests of the components may be combined with the proving ground tests of the cartridge to save expense, upon agreement between the procuring activity and the contractor. In cases where the cartridge specification does not cover all of the proving ground tests specified for the component, the additional tests specified in the component specification shall be conducted.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1310-A017

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